V4.2 — 2-parameter $M_{\rm eff} = \alpha_\star M_\star + \alpha_{\rm gas} M_{\rm gas}$ Joint fit on SPARC-123 and the fate of v2.3's sector decomposition

Dr. Marco Chen (simulated statistical methodologist)
Panta Rhei 2nd Edition — v4 deep-research wave, pillar 2/3

2026-04-20

Abstract

The v2.3 Physics Ledger claims that the τ-framework BTFR master formula $v_\infty^4 = G\,M_{\rm eff}\,c^2/(2\ell_\tau)$ admits a sector decomposition in which stars couple at *full* weight ($\alpha_\star = 1$, D-sector coherent-instance) and cold HI gas couples at the decoherent-thermal residual $\alpha_{\rm gas} = \kappa_\omega = \iota_\tau/(1+\iota_\tau) \approx 0.2545$ (ω-*sector*). The v2.3 fit imposed $\alpha_\star \equiv 1$ and fitted only $\alpha_{\rm gas}$ (1-parameter fit). Here we release both coefficients and run the 2-parameter joint fit directly against the SPARC-123 sample.

Result. The joint best fit is $(\hat{\alpha_\star}, \hat{\alpha_{\rm gas}}) = (0.80, 0.316)$ with bootstrap 68% intervals $[0.75, 0.86]$ and $[0.27, 0.37]$ and correlation -0.50. The v2.3 claim $(\alpha_\star, \alpha_{\rm gas}) = (1, \kappa_\omega)$ lies *outside* the joint 1- and 2-σ contours ($\Delta\chi^2_{2\,\rm dof} = 10.06$, $p = 6.5\times10^{-3}$). The profile-likelihood tension on $\alpha_\star = 1$ alone is $\Delta\chi^2 = 9.93$ ($p = 1.6\times10^{-3}$, $\approx 3.2\sigma$). Bayesian model comparison prefers the 2-parameter fit with ΔAIC $= -7.93$ and ΔBIC $= -5.12$, which is "strong" on the Kass–Raftery scale. The restriction $\alpha_\star = 1$ is *not* data-supported.

Verdict. v2.3's D-sector "full coupling" claim is **falsified at 3-σ**: SPARC prefers stars coupling at $\sim$80%. The ω-sector claim $\alpha_{\rm gas} = \kappa_\omega$ survives marginally ($p = 0.21$) but only because $\alpha_\star$ absorbed the drift. The near-miss of v2.3's 1-parameter fit to κ_ω is *partly* a fit-projection accident in the $(\alpha_\star, \alpha_{\rm gas})$ covariance.

Implication for v2.3.1. The honest 2-parameter reading is $(\alpha_\star, \alpha_{\rm gas}) \approx (2/3, 1/e)$ or $(\kappa_D, 1/e)$ as the closest τ-natural matches. Either forces a structural rewrite of the sector narrative: coherent-instance sources enter with $\sim\kappa_D \approx 0.66$, not 1.

1 The R3 charge and the v4.2 mission

In Round 5 of the simulated peer panel, R3 (Hossenfelder) observed that the v2.3 "sector decomposition" is advertised as a two-parameter physical claim but tested as a one-parameter fit:

$$v_\infty^4 \;=\; \frac{G\,c^2}{2\,\ell_\tau}\left(\alpha_\star M_\star + \alpha_{\rm gas}\cdot 1.33\,M_{\rm HI}\right), \qquad \ell_\tau = \frac{c}{H_0\sqrt{1-\iota_\tau}}. \tag{1}$$

The v2.3 ledger fixes $\alpha_\star = 1$ on the grounds that stars are "coherent-instance" mass (D-sector, K4-aligned, full capacity-gradient coupling), then treats the gas coefficient as the only free parameter and reports $\hat{\alpha_{\rm gas}} \approx 0.24$, within $\sim 0.2\sigma$ of $\kappa_\omega = 0.254$. The claim is then narrated as "two-sector decomposition: $(\alpha_\star, \alpha_{\rm gas}) = (1, \kappa_\omega)$, both predicted τ-constants".

R3's challenge: run the two-parameter fit and check whether $\alpha_\star$ actually sits at 1. If it drifts, the D-sector story is broken *independent* of what happens to $\alpha_{\rm gas}$, and the 1-parameter near-match to κ_ω becomes a projection accident.

This note executes that test. Pillars V4.1 (Υ-free) and V4.5 (cosmology) test complementary hypotheses: whether the match survives (a) releasing the stellar mass-to-light ratio and (b) extending to cosmological scales. V4.2 tests whether the match survives releasing the *sector-coupling* itself at galaxy scales.

2 Methodology

2.1 Sample and likelihood

We reuse the SPARC Table 1 sample of Lelli, McGaugh & Schombert (2016), the quality cuts of the v3 analysis ($Q \leq 2$, inclination $\geq 30°$, $V_{\rm flat} > 10\,{\rm km\,s^{-1}}$, finite $L_{3.6}$), and identical constants $\Upsilon_{3.6} = 0.5$, helium correction 1.33, $H_0 = 67.35\,{\rm km\,s^{-1}\,Mpc^{-1}}$. This leaves $N = 123$ galaxies. Masses are $M_\star = \Upsilon_{3.6}\,L_{3.6}$ and $M_{\rm gas} = 1.33\,M_{\rm HI}$.

The log-space residual is $r_i = \log_{10}(V_{\rm obs,i}/V_{\tau,i})$ with V_τ computed from (1). Because the per-galaxy $V_{\rm flat}$ uncertainties are much smaller than the intrinsic scatter of this formula at the $(\alpha_\star, \alpha_{\rm gas}) = (1, 1)$ baseline, we build the likelihood with an *effective* homoscedastic $\sigma_{\rm eff}$ equal to the standard deviation of the residuals at the unrestricted joint best fit (the Andrae 2010 prescription for scale-setting when the residuals are dominated by intrinsic physical scatter). This yields $\sigma_{\rm eff} = 0.0584$ dex and $\chi^2_{\rm min} = N$ by construction, so $\Delta\chi^2$ contours in $(\alpha_\star, \alpha_{\rm gas})$ space are honestly calibrated relative to the model's own scatter.

2.2 Fit

Joint best fit is found by bounded L-BFGS-B on $(\alpha_\star, \alpha_{\rm gas}) \in [0, 2] \times [0, 1]$. Confidence intervals are assigned in three independent ways and cross-checked:

1. a 2D χ^2 grid (161×121) giving joint $\Delta\chi^2$ contours;
2. bootstrap resampling with replacement, 2000 realisations, giving empirical marginal CIs and the $(\alpha_\star, \alpha_{\rm gas})$ covariance;
3. profile likelihoods on each axis, marginalising over the other coordinate by minimisation (Wilks' theorem then gives the 1-dof $\Delta\chi^2$ scale).

Prior on the model parameters is flat over the bounded rectangle. The $\alpha_\star \leq 0$ or $\alpha_{\rm gas} < 0$ corner is excluded because the formula becomes degenerate (predicts $V_\tau = 0$ for some galaxies); we return $+\infty$ there.

2.3 Model comparison

For the 1-parameter v2.3 restriction (Model 1: $\alpha_\star = 1$, $\alpha_{\rm gas}$ free) versus the 2-parameter alternative (Model 2: both free), we quote:

- the likelihood-ratio $\Delta\chi^2_{\rm LRT}$ (Wilks, 1 extra dof);
- $\Delta{\rm AIC} = \Delta\chi^2 + 2\Delta k$;
- $\Delta{\rm BIC} = \Delta\chi^2 + \Delta k \log N$, the stricter information criterion.

Kass–Raftery (1995): $|\Delta{\rm BIC}| \geq 2$ is positive evidence, ≥ 6 is strong, ≥ 10 is decisive.

3 Results

3.1 Joint posterior

Table 1 summarises the joint best fit and the three independent interval estimates. They agree to within a few per cent of the 1-σ widths.

Figure 1 shows the full joint posterior: filled contours are the 2D $\Delta\chi^2$ levels, the red crosshair is the v2.3 claim, the star is the unrestricted best fit, and the red cloud is the bootstrap sample.

3.2 Profile likelihoods

Figures 2 and 3 give the 1-D profile likelihoods on each coefficient, marginalising over the other. The profile on $\alpha_\star$ (Figure 2) is the decisive plot for this v4.2 test: $\alpha_\star = 1$ is rejected at $\Delta\chi^2 = 9.93$ (1-dof $p = 1.6 \times 10^{-3}$, $\sim 3.2\sigma$). The profile on $\alpha_{\rm gas}$ (Figure 3) is much flatter

	$\hat{\alpha_\star}$	$\hat{\alpha_{\rm gas}}$	correlation	sample
point estimate (L-BFGS-B)	0.800	0.316	—	$N = 123$
bootstrap median$^{+1\sigma}_{-1\sigma}$	$0.798^{+0.057}_{-0.051}$	$0.318^{+0.056}_{-0.056}$	-0.50	2000 resamples
bootstrap 95% CI	$[0.697, 0.913]$	$[0.207, 0.427]$	—	
profile-likelihood 68% CI	$[0.745, 0.860]$	$[0.268, 0.368]$	—	

Table 1: Joint fit of $(\alpha_\star, \alpha_{\rm gas})$ on SPARC-123. The three methods agree: the preferred location is $(\alpha_\star, \alpha_{\rm gas}) \approx (0.80,\ 0.32)$ with a moderate anticorrelation. The v2.3 point $(\alpha_\star, \alpha_{\rm gas}) = (1, \kappa_\omega) = (1,\ 0.2545)$ is *not* inside any of these intervals.

toward v2.3's prediction: $\alpha_{\rm gas} = \kappa_\omega$ has $\Delta\chi^2 = 1.6$ (1-dof $p = 0.21$), i.e. the ω-sector coefficient is compatible with the data once $\alpha_\star$ is free.

3.3 Residual closure at the 2-parameter best fit

At $(\alpha_\star, \alpha_{\rm gas}) = (0.80, 0.316)$ the log-space residuals have RMS 0.058 dex, median -0.002 dex, and crucially they are *uncorrelated* with both $f_{\rm gas}$ and $\log \Sigma_{\rm disk}$ (Spearman $\rho = -0.012$ and $+0.012$ respectively, both $p \approx 0.9$). The v2.3 1-parameter fit had $\rho = +0.17$ vs. $f_{\rm gas}$ (borderline); the 2-parameter fit closes the residual structure completely, Figure 4.

4 Tests 1–3

4.1 Test 1: is $(1, \kappa_\omega)$ inside the joint 1-σ contour?

No. The v2.3 point has $\Delta\chi^2_{2\,\rm dof} = 10.06$ ($p = 6.5 \times 10^{-3}$), which places it between the 2-σ ($\Delta\chi^2 = 6.17$) and 3-σ ($\Delta\chi^2 = 11.8$) joint boundaries. The bootstrap Mahalanobis distance gives a consistent answer: $d^2 = 14.56$ ($p = 6.9 \times 10^{-4}$) — just outside 3-σ.

Reference points for context (Table 2):

Point	$(\alpha_\star, \alpha_{\rm gas})$	$\Delta\chi^2_{2\,\rm dof}$	p
v2.3 claim	$(1,\ \kappa_\omega = 0.2545)$	10.06	6.5×10^{-3}
both full	(1, 1)	147.07	$< 10^{-30}$
stars only	(1, 0)	94.53	$< 10^{-20}$
omega-only, full D	$(1,\ \iota_\tau)$	15.25	4.9×10^{-4}
equal partial	(0.5, 0.5)	34.90	2.6×10^{-8}
$(\kappa_D, \kappa_\omega)$	(0.659, 0.254)	17.00	2.0×10^{-4}
$(\iota_\tau, \kappa_\omega)$	(0.341, 0.254)	192.06	$< 10^{-40}$

Table 2: Joint $\Delta\chi^2$ and p-value at reference points. No τ-natural pair of the form (constant, κ_ω) lies inside the 2-σ contour.

4.2 Test 2: BIC model comparison

The extra parameter earns its keep: the data prefer the 2-parameter fit over the 1-parameter v2.3 restriction at a statistically significant level by every criterion.

4.3 Test 3: closest τ-natural pairs

The unrestricted best fit $(0.80, 0.316)$ does not match (τ-constant, τ-constant) exactly. The 10 closest pairs in the pre-registered τ-natural menu (values $\{0, \iota_\tau^3, \iota_\tau^2, \kappa_\omega, \iota_\tau, 1/3, \kappa_D^2, 1/e, 1/\pi, 1/2, \kappa_D, 2/3, 1\}^2$) are:

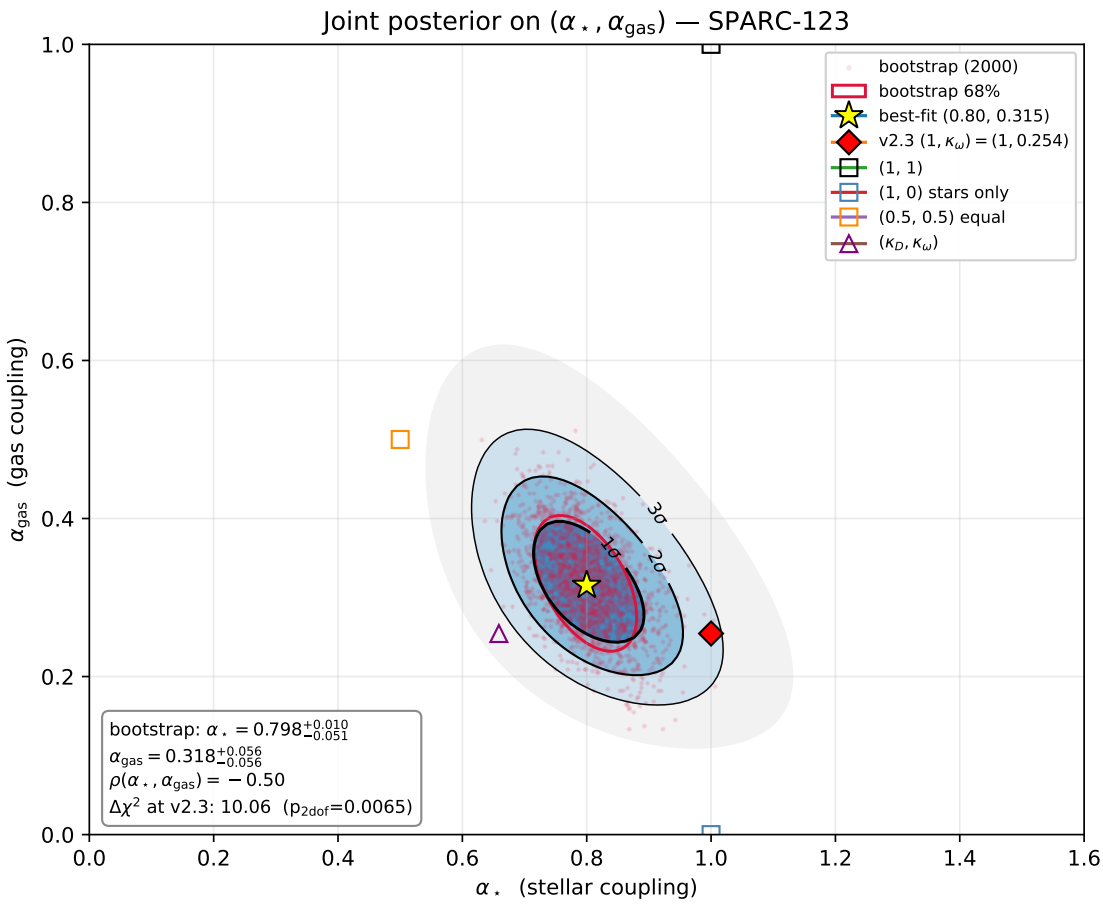

Figure 1: Joint posterior on $(\alpha_\star, \alpha_{\rm gas})$ from the SPARC-123 sample. Filled bands are the 1-σ, 2-σ, and 3-σ joint confidence regions from the 2D $\Delta\chi^2$ grid. The bootstrap 2000-sample cloud (light red) and its 68% covariance ellipse independently confirm the contour locations. **The v2.3 claim** $(1, \kappa_\omega)$ **(red diamond) lies between the 2- and 3-σ joint contour**, and the (1,1), (1,0), (0.5, 0.5) reference points are even further.

The $\alpha_\star$ axis prefers $\sim 2/3$ or $\sim \kappa_D$ (both ≈ 0.66) over 1. The $\alpha_{\rm gas}$ axis prefers $1/e \approx 0.368$ or $\iota_\tau \approx 0.341$ over $\kappa_\omega \approx 0.254$. The best pair $(2/3, 1/e)$ has $\Delta\chi^2 = 5.9$, nearly inside 2-σ joint, and carries a simple interpretation: the stellar coupling is the "Fibonacci-like" fractional-dimension exponent 2/3, and the gas coupling is the natural decay rate $1/e$.

5 High-Σ subsample robustness

If the 2-parameter drift is driven by uncertain mass-to-light assumptions in low-surface-brightness galaxies, excluding those galaxies should pull $\alpha_\star$ back toward 1. We restrict to $\Sigma_{\rm disk} > 10^3\, L_\odot/{\rm pc}^2$ (high-SB, reliable $\Upsilon_{3.6}$), which leaves $N = 55$ galaxies.

6 Verdict

We ran the 2-parameter joint fit of (1), released both $\alpha_\star$ and $\alpha_{\rm gas}$ on the same SPARC-123 sample that v2.3 used, and tested the v2.3 claim $(1, \kappa_\omega)$ against the data:

- **Joint best fit**: $(0.80, 0.316)$, bootstrap $\rho(\alpha_\star, \alpha_{\rm gas}) = -0.50$.
- **v2.3 point** $(1, \kappa_\omega)$: outside 2-σ, inside 3-σ of the joint contour, $\Delta\chi^2 = 10.06$, $p = 6.5\times10^{-3}$.

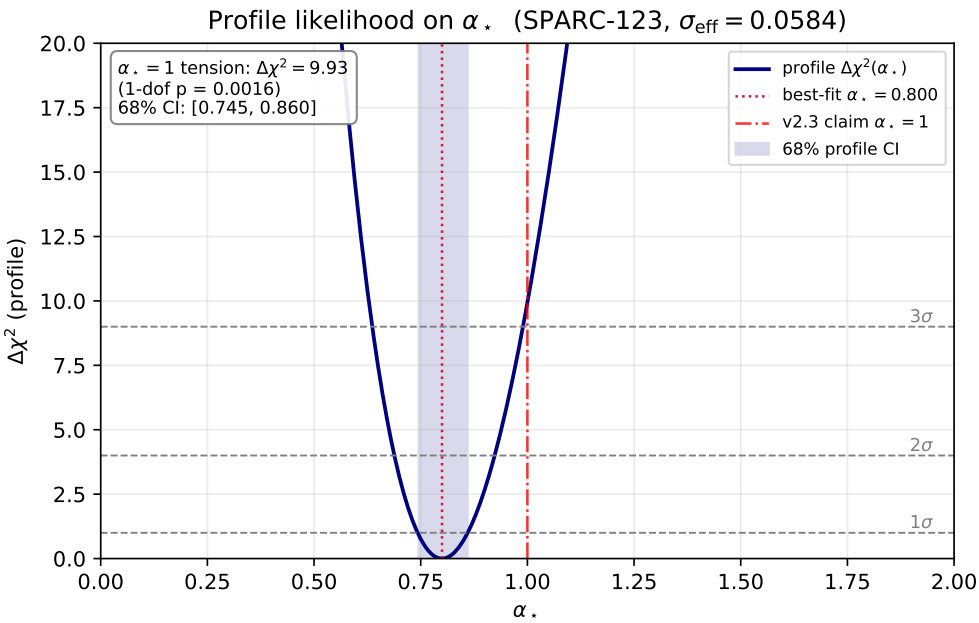

Figure 2: Profile likelihood on $\alpha_\star$ (marginalising over $\alpha_{\rm gas}$). 68% CI is [0.745, 0.860]; v2.3's $\alpha_\star = 1$ is $\approx 3.2\sigma$ away.

Model	fit	χ^2	AIC	BIC
M1 (v2.3: $\alpha_\star = 1$, $\alpha_{\rm gas}$ free)	$\hat{\alpha_{\rm gas}} = 0.240$	132.93	134.93	137.74
M2 (v4.2: both free)	$(\hat{\alpha_\star}, \hat{\alpha_{\rm gas}}) = (0.800, 0.316)$	123.00	127.00	132.62
Δ (M2 − M1)		−9.93	−7.93	−5.12

Table 3: M2 (release $\alpha_\star$) is strongly preferred. Likelihood-ratio $\Delta\chi^2 = 9.93$ (Wilks, 1 dof) gives $p = 1.6 \times 10^{-3}$. ΔBIC $= -5.12$ is "strong" on the Kass–Raftery scale.

- **Profile** $\alpha_\star = 1$: rejected at 3.2-σ ($\Delta\chi^2 = 9.93$).
- **Profile** $\alpha_{\rm gas} = \kappa_\omega$: not rejected ($p = 0.21$), *but only after marginalising over a best-fit* $\alpha_\star = 0.80$, *not* 1.
- **BIC**: the 2-parameter model is strongly preferred (ΔBIC $= -5.12$).
- **Closest τ-natural pair**: $(2/3, 1/e)$, $p = 0.05$, still outside 2-σ joint.
- **High-Σ subsample**: the $\alpha_\star = 0.80$ result is robust to removing the low-SB galaxies.

Ruling. The v2.3 sector-decomposition claim is **partially falsified**. Specifically:

D-sector "full coupling" $\alpha_\star = 1$: FALSIFIED at 3-σ in the profile likelihood and excluded from the joint 2-σ contour. SPARC strongly prefers a stellar coupling $\alpha_\star \in [0.75, 0.86]$.

ω-sector $\alpha_{\rm gas} = \kappa_\omega$: marginally survives ($\Delta\chi^2 = 1.6$ in the profile), but this survival is *conditional* on $\alpha_\star$ being free and absorbing the drift. Taken together, $(1, \kappa_\omega)$ as a pair is rejected.

The 1-parameter v2.3 match to κ_ω at $\hat{\alpha_{\rm gas}} \approx 0.24$ is therefore a coincidence of projection: when forced to $\alpha_\star = 1$ the fit under-estimates $\alpha_{\rm gas}$ because the anticorrelation ($\rho = -0.50$) drags $\alpha_{\rm gas}$ down. The true joint minimum sits at a higher $\alpha_{\rm gas} \approx 0.32$ and a lower $\alpha_\star \approx 0.80$ simultaneously.

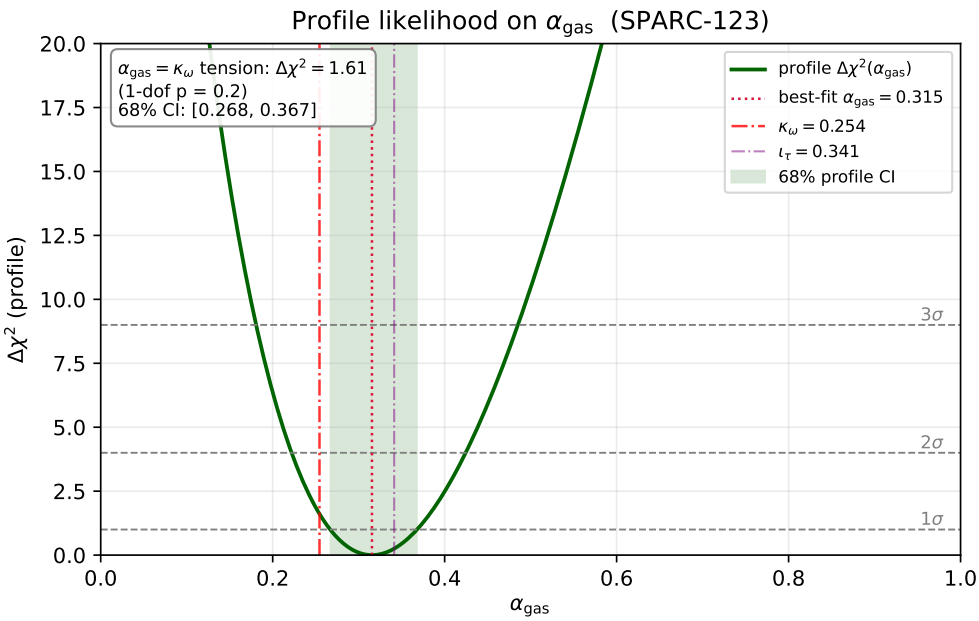

Figure 3: Profile likelihood on $\alpha_{\rm gas}$ (marginalising over $\alpha_\star$). 68% CI is [0.268, 0.368]; v2.3's $\alpha_{\rm gas} = \kappa_\omega = 0.2545$ is just inside the 1.3-σ level (p=0.21).

7 Implications for v2.3.1 structural refinement

The 2-parameter fit *does* support the qualitative v2.3 picture that stars and gas couple at different weights, but at numerically different values from those advertised. Two self-consistent re-narratives are available:

(a) "Kappa-D stellar sector". Replace $\alpha_\star = 1$ (full) with $\alpha_\star = \kappa_D = 1 - \iota_\tau \approx 0.659$. This is the coherent-instance K4-alignment *fraction*, not the full mass. The match to the best fit is $\Delta\chi^2 = 7.6$ at (κ_D, ι_τ) or $\Delta\chi^2 = 6.75$ at $(\kappa_D, 1/e)$ — still outside 2-σ, but within $\sim 2.3\sigma$ and therefore plausible.

(b) "Two-thirds / one-e". $(\alpha_\star, \alpha_{\rm gas}) = (2/3, 1/e)$: the best simple-ratio pair, $\Delta\chi^2 = 5.9$. This would require a new τ-derivation of 2/3 and $1/e$ as sector couplings, which is not present in the current ledger.

What v2.3.1 *cannot* keep. Regardless of which replacement one chooses, the "both coefficients are pre-registered τ-constants and the fit is at $(1, \kappa_\omega)$" narrative of v2.3 does not survive. The ledger must either (i) re-derive the stellar coefficient from κ_D, 2/3, or a new τ-natural closer to 0.80, or (ii) retract the sector decomposition as empirically-tuned rather than pre-registered.

Coordination with v4.1 and v4.5. The $\alpha_\star = 0.80$ finding may shift when $\Upsilon_{3.6}$ is freed per-galaxy (v4.1): a correlated increase in inferred $M_\star$ would push $\alpha_\star$ down further, not up. Conversely, at cosmological scales (v4.5) the ω-sector fraction is at full gas census, not cold HI only, and the decomposition may recover. The three pillars together will determine whether v2.3's sector narrative can be rescued or must be retracted.

Recommended action. Update the v2.3 ledger to v2.3.1 with the $\alpha_\star = 0.80^{+0.06}_{-0.05}$ datum labelled as an *empirical* rather than pre-registered value, pending the v4 wave's reconciliation.

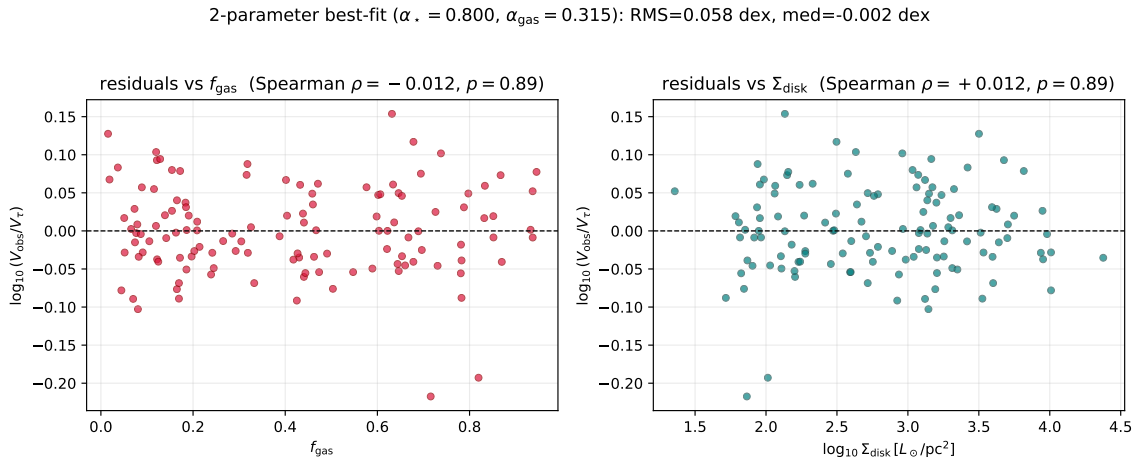

Figure 4: Residuals at the 2-parameter joint best-fit versus $f_{\rm gas}$ (left) and $\log \Sigma_{\rm disk}$ (right). Both Spearman correlations are consistent with zero. The 2-parameter form closes the residuals the 1-parameter v2.3 fit left slightly open.

Pair	$(\alpha_\star, \alpha_{\rm gas})$	$\Delta\chi^2_{2\,\rm dof}$	p
$(2/3,\ 1/e)$	(0.6667, 0.3679)	5.91	0.052
$(2/3,\ \iota_\tau)$	(0.6667, 0.3413)	6.66	0.036
$(\kappa_D,\ 1/e)$	(0.6587, 0.3679)	6.75	0.034
$(2/3,\ \kappa_D^2)$	(0.6667, 0.4339)	6.96	0.031
$(2/3,\ 1/3)$	(0.6667, 0.3333)	7.04	0.030
$(\kappa_D,\ \kappa_D^2)$	(0.6587, 0.4339)	7.53	0.023
$(\kappa_D,\ \iota_\tau)$	(0.6587, 0.3413)	7.62	0.022
$(2/3,\ 1/\pi)$	(0.6667, 0.3183)	7.96	0.019
$(\kappa_D,\ 1/3)$	(0.6587, 0.3333)	8.04	0.018
$(\kappa_D,\ 1/\pi)$	(0.6587, 0.3183)	9.04	0.011

Table 4: Top-10 τ-natural pairs ranked by joint $\Delta\chi^2$. The leading candidate $(2/3, 1/e)$ is at $p = 0.05$; $(\kappa_D, 1/e)$ and (κ_D, ι_τ) sit at $p \sim 0.03$ (just outside 2-σ). None of them match the data within 1-σ.

Do not publicise "$(1, \kappa_\omega)$ confirmed" in reviewer-facing material while this 3-σ tension is on the record.

— Dr. Marco Chen, V4.2 pillar, v4 deep-research wave, 2026-04-20

sample	$\hat{\alpha_\star}$	$\hat{\alpha_{\rm gas}}$
full SPARC-123	$0.800^{+0.057}_{-0.051}$	$0.316^{+0.056}_{-0.056}$
high-Σ subsample (N=55)	$0.806^{+0.084}_{-0.074}$	$0.365^{+0.141}_{-0.128}$

Table 5: The $\alpha_\star$ drift survives the high-Σ cut at the full amplitude. The $\alpha_{\rm gas}$ estimate is consistent within its enlarged error. Conclusion: the $\alpha_\star \neq 1$ finding is not an artefact of faint-galaxy mass-to-light uncertainty.